

DATA GOVERNANCE PORTFOLIO

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This portfolio presents governance artefacts built across enterprise data modernisation projects spanning HR, Finance, Sales, and Marketing domains. All data is anonymised sample data created to demonstrate real-world governance practice.

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01 — Data Catalogue

Enterprise asset inventory covering 12 source systems across 4 business units and 1 group company

The Data Catalogue provides a centralised inventory of all data assets, enabling discoverability, ownership accountability, and lineage traceability across the organisation.

Asset ID	Data Asset Name	Domain	Source System	Owner	Classification	Update Freq	Status
DA-001	Employee Master	HR	HR ERP	HR Director	PII — Restricted	Real-time	Active
DA-002	Payroll Transactions	HR	Payroll ERP	CFO	Financial — Restricted	Monthly	Active
DA-003	Leave & Attendance	HR	HRMS	HR Manager	PII — Internal	Daily	Active
DA-004	General Ledger	Finance	Finance ERP	CFO	Financial — Restricted	Daily	Active
DA-005	Cost Centre Master	Finance	Finance ERP	Finance Manager	Internal	Monthly	Active
DA-006	Budget vs Actuals	Finance	Finance ERP	CFO	Financial — Restricted	Monthly	Active
DA-007	Sales Transactions	Sales	CRM	Sales Director	Internal	Real-time	Active
DA-008	Customer Master	Sales	CRM	Sales Director	PII — Restricted	Daily	Active
DA-009	Sales Pipeline	Sales	CRM	Sales Manager	Internal	Daily	Active
DA-010	Campaign Data	Marketing	Marketing Platform	CMO	Internal	Weekly	Active
DA-011	Lead Generation	Marketing	Marketing Platform	CMO	PII — Restricted	Daily	Active
DA-012	NPS & Customer Feedback	Marketing	Survey Tool	CMO	PII — Restricted	Monthly	Active

Classification Legend: PII — Restricted (personal data, access controlled) | Financial — Restricted (sensitive financial data) | Internal (business use only)

02 — Data Dictionary

Field-level metadata definitions — HR & Payroll Domain (DA-001, DA-002)

The Data Dictionary defines field-level metadata including business definitions, data types, allowed values, transformation rules, and KPI linkages. Maintained as the single source of truth for Data Engineering and BI Developer teams.

Field Name	Table	Data Type	Business Definition	Allowed Values	Nullabl e	Transformation Rule	KPI Link
employee_id	dim_employee	VARCHAR(10)	Unique identifier per active employee	Non-null, alphanumeric, EMP+7 digits	No	Source as-is from HR ERP	Headcount
full_name	dim_employee	VARCHAR(100)	Legal full name of employee	Non-null, alpha + spaces	No	CONCAT(first_name, ' ', last_name)	—
department	dim_employee	VARCHAR(50)	Business unit the employee belongs to	HR, Finance, Sales, Marketing, IT	No	Lookup from dept_master table	Dept Headcount
hire_date	dim_employee	DATE	Date employee joined the organisation	YYYY-MM-DD, not future date	No	FORMAT source date to YYYY-MM-DD	Tenure KPI
employment_type	dim_employee	VARCHAR(20)	Nature of employment contract	Full-Time, Part-Time, Contract	No	Source as-is, validated against enum	FTE Ratio
is_active	dim_employee	BOOLEAN	Flag indicating current employment status	TRUE / FALSE	No	TRUE if termination_date IS NULL	Active Headcount
termination_date	dim_employee	DATE	Date employee exited the organisation	YYYY-MM-DD or NULL if active	Yes	NULL for active employees	Attrition Rate
gross_salary	fact_payroll	DECIMAL(12,2)	Total pre-deduction salary for the pay period	> 0, numeric	No	SUM of all salary components	Payroll Cost
deductions	fact_payroll	DECIMAL(12,2)	Total statutory and voluntary deductions	>= 0, numeric	No	SUM(tax + pf + other_deductions)	Net Pay Ratio
net_pay	fact_payroll	DECIMAL(12,2)	Take-home salary after all deductions	> 0, numeric	No	gross_salary - deductions	Net Pay Ratio
pay_period	fact_payroll	DATE	Month and year of payroll run	First day of month, YYYY-MM-01	No	TRUNC(pay_date, 'MM')	Monthly Payroll
cost_centre	fact_payroll	VARCHAR(10)	Organisational unit charged for payroll cost	Valid codes from cost_centre_master	No	FK to cost_centre_master	Cost per Dept

All field definitions reviewed and approved by the Data Governance team. Any changes to field definitions must be raised via Azure DevOps change request.

03 — Data Lineage Document

End-to-end data flow — Employee Headcount & Payroll Cost Dashboard

Data lineage traces the complete journey of data from source systems through transformation layers to final BI consumption. This document covers the HR domain lineage for the Headcount & Payroll Cost Dashboard.

Layer	System / Object	Type	Description	Ingestion Method	Responsible Team
Source	HR ERP (Employee Master)	Operational DB	Raw employee records including personal, contract and department data	API Extract / Daily	Data Engineering
Source	Payroll ERP (Payroll Transactions)	Operational DB	Monthly payroll run data — gross, deductions, net pay per employee	Flat File / Monthly	Data Engineering
Raw Layer	raw.employees raw.payroll_txn	Snowflake Table	As-is landing zone — no transformations, full history retained	dbt source	Data Engineering
Staging Layer	stg_employees stg_payroll	dbt Model	Cleaned, typed, renamed fields. Nulls handled. PII masked where required	dbt run	Data Engineering
Mart Layer	dim_employee fact_payroll	dbt Model	Conformed dimensions and facts. Business rules applied. KPIs calculated	dbt run	Data Engineering
Semantic Layer	Power BI Dataset	BI Model	DAX measures, relationships, row-level security applied for business users	Power BI Refresh / Daily	BI Developers
Consumption	Headcount & Payroll Cost Dashboard	Power BI Report	Final dashboard consumed by HR Director, CFO, and Department Heads	Live / Scheduled Refresh	BI Developers

Business Rules Applied at Mart Layer

Rule ID	Rule Name	Logic
BR-001	Active Employee Filter	is_active = TRUE where termination_date IS NULL
BR-002	Headcount Calculation	COUNT(DISTINCT employee_id) WHERE is_active = TRUE
BR-003	Attrition Rate	(COUNT(terminations in period) / AVG(headcount)) * 100
BR-004	Payroll Cost Ratio	(SUM(gross_salary) / SUM(total_revenue)) * 100
BR-005	Net Pay Ratio	SUM(net_pay) / SUM(gross_salary) * 100
BR-006	Cost per Department	SUM(gross_salary) GROUP BY cost_centre

04 — KPI Governance Framework

Standardised KPI definitions across HR, Finance, Sales, and Marketing domains

The KPI Governance Framework defines, standardises, and governs all metrics used across the organisation. Each KPI has a single agreed definition, formula, owner, and refresh schedule — eliminating conflicting numbers across departments and ensuring consistent reporting.

KPI Name	Domain	Definition	Formula	Data Source	Owner	Refresh	Threshold	D
Active Headcount	HR	Total employees on active payroll at period end	COUNT(employee_id) WHERE is_active=TRUE	HR ERP	HR Director	Daily	Target: Plan	HR
Attrition Rate	HR	% of employees who left during the period	(Exits / Avg Headcount) x 100	HR ERP	HR Director	Monthly	>15% = Red	HR
Average Tenure	HR	Average years of service across active employees	AVG(TODAY - hire_date) in years	HR ERP	HR Manager	Monthly	< 2yr = Amber	HR
FTE Ratio	HR	Ratio of full-time to total headcount	Full-Time Count / Total Headcount x 100	HR ERP	HR Director	Monthly	< 70% = Amber	HR
Payroll Cost Ratio	Finance	Payroll as a % of total revenue	SUM(gross_salary) / Total Revenue x 100	Finance ERP	CFO	Monthly	>40% = Red	Finance
Budget Variance	Finance	Difference between actual spend and approved budget	Actual - Budget / Budget x 100	Finance ERP	CFO	Monthly	>5% = Amber	Finance
Cost per Employee	Finance	Average total cost per active employee per month	Total Payroll Cost / Active Headcount	Finance ERP	Finance Mgr	Monthly	MoM trend	Finance
Revenue per Employee	Finance	Revenue generated per active employee	Total Revenue / Active Headcount	Finance + HR	CFO	Monthly	YoY growth	Finance
Sales Revenue	Sales	Total confirmed revenue from closed deals in period	SUM(deal_value) WHERE status='Closed Won'	CRM	Sales Director	Daily	vs Target	Sales
Win Rate	Sales	% of deals closed won vs total deals in final stage	Closed Won / (Closed Won + Closed Lost) x 100	CRM	Sales Director	Weekly	< 30% = Red	Sales
Pipeline Coverage	Sales	Ratio of pipeline value to revenue target — measures forecast health	Total Open Pipeline Value / Revenue Target	CRM	Sales Manager	Weekly	< 3x = Amber	Sales
Avg Deal Cycle Time	Sales	Average days from opportunity creation to deal closure	AVG(close_date - created_date) in days	CRM	Sales Manager	Monthly	> 60d = Amber	Sales
Lead Conversion Rate	Marketing	% of marketing leads that convert to qualified sales opportunities	Converted Leads / Total Leads x 100	Marketing + CRM	CMO	Weekly	< 10% = Red	Marketing
Cost per Lead	Marketing	Average marketing spend required to generate one lead	Total Campaign Spend / Total Leads Generated	Marketing	CMO	Monthly	MoM trend	Marketing
Campaign ROI	Marketing	Return on marketing investment — revenue attributed vs spend	(Attributed Revenue - Spend) / Spend x 100	Marketing + CRM	CMO	Monthly	< 200% = Amber	Marketing
NPS Score	Marketing	Net Promoter Score — likelihood of customers recommending the organisation	% Promoters - % Detractors	Survey Tool	CMO	Quarterly	< 30 = Amber	Marketing

05 — Source-to-Target Mapping

Field-level mapping from source systems to Snowflake warehouse — HR & Finance domains

Source-to-Target Mapping documents the exact field-level transformation from each source system to the Snowflake data warehouse. This serves as the contract between Business, Data Engineering, and BI teams — ensuring no ambiguity in how data is moved and transformed.

Source System	Source Table	Source Field	Src Type	Target Table	Target Field	Tgt Type	Transformation Rule	PK/FK	Notes
HR	EMPLOYEE	EMP_CODE	VARCHAR	dim_employee	employee_id	VARCHAR(10)	Direct map	PK	Primary key for dim_employee
HR	EMPLOYEE	FIRST_NM	VARCHAR	dim_employee	full_name	VARCHAR(100)	CONCAT(FIRST_NM, ' ', LAST_NM)	—	Full name derived from first and last name
HR	EMPLOYEE	DEPT_CD	VARCHAR	dim_employee	department	VARCHAR(50)	Lookup: dept_master.dept_name	FK	Foreign key to dept_master
HR	EMPLOYEE	JOIN_DT	DATE	dim_employee	hire_date	DATE	FORMAT to YYYY-MM-DD	—	Standard date format
HR	EMPLOYEE	EMP_TYPE	VARCHAR	dim_employee	employment_type	VARCHAR(20)	CASE: FT=Full-Time, PT=Part-Time	—	Standard employment types
HR	EMPLOYEE	TERM_DT	DATE	dim_employee	termination_date	DATE	NULL if no termination	—	Termination date or NULL
HR	EMPLOYEE	TERM_DT	DATE	dim_employee	is_active	BOOLEAN	TRUE if TERM_DT IS NULL	—	Active status based on termination date
ERP	PAY_TXN	EMPL_ID	VARCHAR	fact_payroll	employee_id	VARCHAR(10)	FK to dim_employee.employee_id	FK	Foreign key to dim_employee
ERP	PAY_TXN	PAY_DT	DATE	fact_payroll	pay_period	DATE	TRUNC(PAY_DT, 'MM') — 1st of month	—	Pay period start date
ERP	PAY_TXN	GROSS_AMT	NUMBER	fact_payroll	gross_salary	DECIMAL(12, 2)	Direct map, validated > 0	—	Gross salary amount
ERP	PAY_TXN	DEDN_AMT	NUMBER	fact_payroll	deductions	DECIMAL(12, 2)	SUM(TAX+PF+OTHER_DEDN)	—	Total deductions
ERP	PAY_TXN	NET_AMT	NUMBER	fact_payroll	net_pay	DECIMAL(12, 2)	GROSS_AMT - DEDN_AMT	—	Net pay amount
ERP	GL_ENTRIES	CC_CODE	VARCHAR	dim_cost_centre	cost_centre_id	VARCHAR(10)	Direct map	PK	Primary key for dim_cost_centre
ERP	GL_ENTRIES	CC_NAME	VARCHAR	dim_cost_centre	cost_centre_name	VARCHAR(100)	Direct map	—	Cost centre name
ERP	BUDGET	BUDGET_AMT	NUMBER	fact_budget	budget_amount	DECIMAL(14, 2)	Direct map, validated > 0	—	Budget amount
CRM	OPPORTUNITIES	OPP_ID	VARCHAR	fact_sales	opportunity_id	VARCHAR(20)	Direct map	PK	Primary key for fact_sales
CRM	OPPORTUNITIES	OPP_VALUE	NUMBER	fact_sales	deal_value	DECIMAL(14, 2)	Direct map, currency normalised GBP	—	Deal value in GBP
CRM	OPPORTUNITIES	STAGE	VARCHAR	fact_sales	deal_status	VARCHAR(30)	Validated against stage_enum table	—	Deal status

06 — Data Quality Scorecard

Quality measurement across 5 dimensions for key enterprise datasets

The Data Quality Scorecard measures and monitors data quality across five dimensions: Completeness, Accuracy, Consistency, Timeliness, and Uniqueness. RAG status triggers remediation actions managed through Azure DevOps.

Dataset	Domain	Completeness	Accuracy	Consistency	Timeliness	Uniqueness	Overall Score	RAG	Action Required
Employee	HR	99.2%	98.7%	97.5%	100%	100%	99.1%	● Green	None
Payroll	HR	98.5%	97.2%	96.8%	98.0%	100%	98.1%	● Green	None
Cost_centre	Finance	100%	99.5%	99.0%	100%	100%	99.7%	● Green	None
Budget	Finance	95.0%	93.4%	91.2%	85.0%	100%	92.9%	● Amber	Budget upload delay — DR raised
Sales	Sales	97.8%	96.5%	95.0%	99.0%	99.5%	97.6%	● Green	None
Customer	Sales	89.3%	87.6%	84.2%	100%	96.0%	91.4%	● Amber	Missing email/phone — DR enrichment in progress
Campaign	Marketing	94.5%	92.0%	90.5%	96.0%	98.0%	94.2%	● Green	None
Leads	Marketing	78.5%	75.3%	72.0%	90.0%	94.0%	81.9%	● Red	Source CRM sync failure — ticket open
Prospects	Marketing	99.0%	98.5%	98.0%	92.0%	100%	97.5%	● Green	None

Data Quality Dimension Definitions

Dimension	Definition	Measurement Method	Acceptable Threshold
Completeness	% of required fields populated with non-null values	NULL count / Total records	>= 95%
Accuracy	% of records that correctly reflect real-world values	Validation rules + source reconciliation	>= 95%
Consistency	Same values across systems and datasets for the same entity	Cross-system comparison checks	>= 95%
Timeliness	Data available within agreed SLA from source	Load timestamp vs SLA window	>= 95%
Uniqueness	Absence of duplicate records for primary key fields	Duplicate key count / Total records	>= 98%

07 — Business Glossary

Standardised definitions of business terms across HR, Finance, Sales, and Marketing domains

The Business Glossary is the authoritative reference for all business terminology used across dashboards, reports, and data products. It eliminates ambiguity by ensuring every term has one agreed definition, owner, and domain classification.

Term ID	Term	Domain	Definition	Related Terms	Owner	Last Updated
BG-001	Active Employee	HR	An employee currently on payroll with no termination date recorded in the HR system	Headcount, FTE	HR Director	Jun 2026
BG-002	Headcount	HR	Total count of active employees at a given point in time, including full-time, part-time, and contract staff	FTE, Active Employee	HR Director	Jun 2026
BG-003	FTE (Full-Time Equiv.)	HR	Standardised unit of employee capacity. One FTE = one full-time employee. Part-time = 0.5 FTE	Headcount, Part-Time	HR Manager	Jun 2026
BG-004	Attrition	HR	Voluntary or involuntary departure of an employee from the organisation during a defined period	Retention Rate, Turnover	HR Director	Jun 2026
BG-005	Attrition Rate	HR	Percentage of employees who left the organisation relative to average headcount in the same period	Attrition, Headcount	HR Director	Jun 2026
BG-006	Tenure	HR	Length of time an employee has been continuously employed by the organisation, measured from hire date	Hire Date, Active Employee	HR Manager	Jun 2026
BG-007	Gross Salary	Finance	Total pre-deduction compensation payable to an employee for a pay period, including base pay and allowances	Net Pay, Deductions	CFO	Jun 2026
BG-008	Net Pay	Finance	Take-home salary after all statutory and voluntary deductions have been applied to gross salary	Gross Salary, Deductions	CFO	Jun 2026
BG-009	Cost Centre	Finance	An organisational unit to which costs are allocated for management accounting and budget tracking purposes	Budget, Department	Finance Manager	Jun 2026
BG-010	Budget Variance	Finance	The difference between actual expenditure and approved budget for a given period, expressed as % of budget	Budget, Actuals	CFO	Jun 2026
BG-011	Payroll Cost Ratio	Finance	Total payroll cost expressed as a percentage of total revenue, used to monitor workforce cost efficiency	Gross Salary, Revenue	CFO	Jun 2026
BG-012	Revenue per Employee	Finance	Total organisational revenue divided by active headcount — measures productivity per employee	Revenue, Headcount	CFO	Jun 2026
BG-013	Closed Won	Sales	A sales opportunity where the prospect has signed a contract or committed to purchase — counted as realised revenue	Pipeline, Deal Value	Sales Director	Jun 2026
BG-014	Pipeline	Sales	Total value of all open sales opportunities currently being pursued, regardless of probability of closure	Closed Won, Win Rate	Sales Director	Jun 2026

Term ID	Term	Domain	Definition	Related Terms	Owner	Last Updated
BG-015	Win Rate	Sales	Percentage of deals reaching final stage that are closed as won, measured over a defined period	Pipeline, Closed Won	Sales Manager	Jun 2026
BG-016	Deal Cycle Time	Sales	Average number of days from opportunity creation to deal closure (won or lost)	Pipeline, Closed Won	Sales Manager	Jun 2026
BG-017	Lead	Marketing	A prospective customer who has shown interest through a marketing interaction and has been captured in the CRM	Conversion Rate, MQL	CMO	Jun 2026
BG-018	MQL	Marketing	Marketing Qualified Lead — a lead that has met predefined engagement criteria and is ready for sales follow-up	Lead, SQL	CMO	Jun 2026
BG-019	Lead Conversion Rate	Marketing	Percentage of marketing leads that progress to a qualified sales opportunity within a defined time window	Lead, MQL, Pipeline	CMO	Jun 2026
BG-020	Campaign ROI	Marketing	Return on marketing investment — revenue attributed to a campaign minus campaign spend, as % of spend	Cost per Lead, Revenue	CMO	Jun 2026
BG-021	NPS (Net Promoter Score)	Marketing	Customer loyalty metric: % of promoters minus % of detractors from a standardised 0-10 recommendation survey	CSAT, Customer Feedback	CMO	Jun 2026
BG-022	Cost per Lead	Marketing	Total marketing spend in a period divided by total leads generated — measures campaign efficiency	Campaign ROI, Lead	CMO	Jun 2026
BG-023	Data Steward	Governance	The person responsible for the day-to-day management of a data asset — ensuring quality, definitions, and standards are maintained	Data Owner, DG	Data Mgmt Team	Jun 2026
BG-024	Data Owner	Governance	The business executive accountable for a data domain — approves definitions, access policies, and quality standards	Data Steward	Exec Team	Jun 2026
BG-025	Golden Record	Governance	The single, authoritative, trusted version of a data entity (e.g. one customer record) after deduplication and master data management	MDM, Deduplication	Data Mgmt Team	Jun 2026

All terms reviewed and approved by domain owners. Change requests to existing definitions must be raised via the Data Governance change process and approved by the relevant domain owner before publication.